

Hang Gao

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SUMMARY

My research spans large language models, generative AI, and machine learning systems, with work in retrieval-augmented generation, information retrieval, Agentic AI, and multimodal reasoning. I have 4+ years of experience developing scalable GenAI and RAG systems from research ideation through prototyping, evaluation, and implementation, with strong Python and PyTorch engineering skills in LLM training and evaluation, Agentic AI systems, and end-to-end GenAI applications. My research interests center on how models retrieve, represent, and reason over complex information, and how to make these systems more reliable, efficient, and scalable.

EDUCATION

Rutgers University-New Brunswick, Piscataway, NJ, US Sep 2022 – Expected May 2027

Ph.D. Candidate in Computer Science, advised by Dimitris N. Metaxas

Syracuse University, Syracuse, NY, US Sep 2020 – May 2022

M.S. in Computer Science

Jilin University, Changchun, China Sep 2016 – Jul 2020

B.Eng. in Logistics Engineering

EXPERIENCE

Syracuse University — *Research Assistant* June 2021 – May 2022

Fairness in Social Network.

Syracuse, NY

(Advisor: Sucheta Soundarajan)

- Developed graph sparsification algorithms to improve fairness and interpretability in social network analysis while preserving key structural properties.
- Reduced community expansion discrepancy by 18% using an optimized edge-weighting and deletion strategy on complex network data.

NEC Laboratories America — *Research Intern* May 2025 – Aug 2025

Scalable GraphRAG System for Domain-Specific Enterprise QA

Princeton, NJ

(Mentors: Yangmin Ding & Shaobo Han)

- Built an end-to-end GraphRAG research prototype for domain-specific question answering, organizing documents into chunk–entity–concept bipartite graphs to reduce hallucinated entity–entity relations. (*IJCNLP–AAACL 2026*)
- Designed and implemented a concept-mediated retrieval pipeline that maps user queries to entities and concepts, expands semantically related evidence, and reranks chunks for reliable multi-hop question answering.
- Implemented a localized graph update workflow that incorporated newly ingested documents without reconstructing the full graph, supporting continuously evolving document collections.

Rutgers University-New Brunswick — *Research Assistant* Sep 2023 – Present

Grounding Foundation Models: From Retrieval to Reliable Reasoning

New Brunswick, NJ

(Advisor: Dimitris N. Metaxas)

- Formulated VRSD, a parameter-free set-level retrieval optimization objective that balances similarity and diversity, proving NP-completeness and developing an efficient relevance-aware diversity heuristic for embedding selection.
- Built INSES, an agentic GraphRAG framework over noisy and sparse knowledge graphs, integrating semantic pruning, embedding-based expansion, and cost-aware routing to improve robust reasoning beyond explicit edges. (*ICML 2026*)
- Developed CLIMB, a training-free multimodal framework that constructs complementary evidence pools and uses confidence-controlled refinement for knowledge-intensive visual question answering. (*EMNLP 2026 Findings*)
- Characterized Semantic Shift as a source of long-text embedding collapse and analyzed how content transitions and pooling distort document representations and degrade retrieval quality.

SELECTED PUBLICATIONS

- [1] **Hang Gao**, Dimitris N. Metaxas. “Betting on Truth: Shaping LLM Metacognitive Confidence through Game-Theoretic Incentives and Batch Contrast.”
- [2] **Hang Gao**, Wujiang Xu, Kai Mei, Dimitris N. Metaxas. “Pooling and Semantic Shift: The Fundamental Challenges in Long Text Embedding and Retrieval.” [\[PDF\]](#)
- [3] **Hang Gao**, Wujiang Xu, Zhixing Zhang, Kai Mei, Jingyi Yang, Dimitris N. Metaxas. “CLIMB: Confidence-Guided Complementary Evidence for Multimodal Retrieval-Augmented Generation.” (*EMNLP 2026 Findings*)
- [4] **Hang Gao**, Dimitris N. Metaxas. “Beyond Explicit Edges: Robust Reasoning over Noisy and Sparse Knowledge Graphs.” (*ICML 2026*) [\[PDF\]](#)
- [5] **Hang Gao**, Yangmin Ding, Shaobo Han, Zhuocheng Jiang. “BG-RAG: Concept-Mediated Bipartite Graphs for Scalable RAG.” (*IJCNLP-AACL 2026*)
- [6] **Hang Gao**, Yongfeng Zhang. “INMS: Memory Sharing for Large Language Model-Based Agents.” (*IJCNLP-AACL 2026*) [\[PDF\]](#)
- [7] **Hang Gao**, Yongfeng Zhang. “Task-Aligned Tool Recommendation for Large Language Models.” (*IJCNLP-AACL 2025*) [\[PDF\]](#)
- [8] **Hang Gao**, Dong Deng, Yongfeng Zhang. “Vector Retrieval with Similarity and Diversity: How Hard Is It?” [\[PDF\]](#)
- [9] Wujiang Xu, Zujie Liang, Kai Mei, **Hang Gao**, Juntao Tan, Yongfeng Zhang. “A-Mem: Agentic Memory for LLM Agents.” (*NeurIPS 2025*) [\[PDF\]](#)

TECHNICAL SKILLS

- **Programming:** Python, Java
- **Machine Learning:** PyTorch, Transformers, LLMs, agentic AI, NumPy, RAG, AI agents, GenAI
- **Tools:** LlamaIndex, LangChain, Linux, Docker, L^AT_EX, NetworkX, FAISS

ACADEMIC SERVICE

Conference Reviewer

- International Conference on Learning Representations (ICLR)
- International Conference on Machine Learning (ICML)
- AAAI Conference on Artificial Intelligence (AAAI)
- Conference on Neural Information Processing Systems (NeurIPS)

TEACHING

CS 214 – Systems Programming	Spring 2024
CS 512 – Introduction to Data Structures and Algorithms	Fall 2023
CS 509 – Foundations of Computer Science	Spring 2023
CS 416 – Operating Systems Design	Fall 2022